Report No. 2
Team sySTEMatic
02/02/2026

January Report - Team sySTEMatic

1. Introduction

During this sprint, we have focused on software stack improvements, including architectural changes and quality-of-life additions. An important improvement is the implementation of improved traffic sign handling logic.

2. Implemented improvements

Architectural

- separation of different software components of the system into separate files (AI engine, web server, PID control system);
- implementation of an unified configuration file that controls configuration parameters across all modules;
- implementation of **multiprocessing**, each module is launched as a separate process, and Queues are used to send data between processes.

Quality-of-life

- added video recording system that facilitates recording videos for AI training data;
- added **emergency stop** button in the Web UI;
- general Web UI improvements.

Autonomous driving

- improved sign handling logic (for now just for the STOP sign).

  The system considers the **number of frames** since the sign started being detected by the AI engine while also calculating the **surface area** of the sign, thus detecting how far is it from the vehicle, ensuring the correct triggering of the braking system (in the case of the STOP sign).

3. General status of the project

The system uses an AI engine powered by the YOLO11n segmentation model, running at 320x320 resolution, which is currently capable of detecting 14 classes.

For lane detection, the system extracts the geometric centroid of the segmentation mask, then calculates the lateral error from the image center, driving a PID controller to compute the steering angle.

The system is able to correctly handle a STOP sign and stop in time, as explained in the Improvements section.

The overall status of the project is that the car can maintain the lane in most conditions, can handle STOP signs, can recognize but not handle other traffic signs. The car may lose lane tracking in poor lighting conditions, due to lack of AI training on our own race track.

4. Upcoming activities

We plan to improve our AI engine, by switching to the latest YOLO26 model, optimized for edge AI. Besides switching the model architecture, we want to generate training data for our own race track, experiment with AI augmentation and hyperparameter tuning.

We also want to experiment with adjusting the camera handling, as currently the system crops the image to 320px, resulting in a reduced field of view (FOV). By downscaling the image instead of cropping, we can take full advantage of the camera and its high FOV.